Práctica 2: Sistema Cardiovascular

Departamento de Ingeniería Eléctrica y Electrónica, Ingeniería Biomédica

Tecnológico Nacional de México [TecNM - Tijuana], Blvd. Alberto Limón Padilla s/n, C.P. 22454, Tijuana, B.C., México

Table of Contents

Información general....1
Datos de la simulación....1
1. Respuestas: Hipotenso, Normotenso e Hipertenso....2
2. Respuestas: Normotenso, Hipotenso y Controlador....2
3. Respuestas: Normotenso, Hipertenso y Controlador....3
Modelo de EDOs....3
Funcion: Respuestas de simulacion....4

Información general



Nombre del alumno: Kevin Rodriguez Padilla

Número de control: 22210428

Correo institucional: l22210428**@tectijuana.edu.mx**

Asignatura: Modelo de Sistema Cardiovascular

Docente: **Dr. Paul Antonio Valle Trujillo; paul.valle@tectijuana.edu.mx**

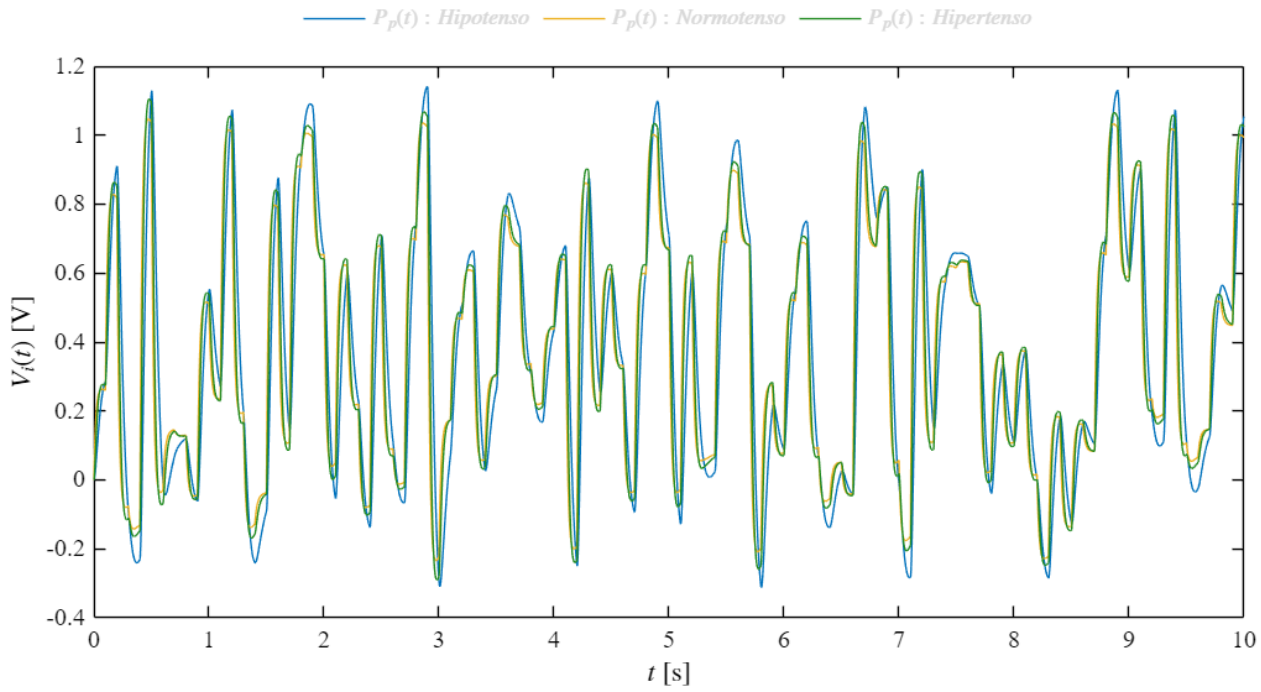
Datos de la simulación

```
clc; clear; close all; warning('off','all')
open_system('system')
parameters.StopTime = '15';
parameters.Solver = 'ode15s';
parameters.MaxStep = '1e-3';
```

```
x = sim('system',parameters);
t_data = (0:0.1:15)';
t = (0:0.001:15)';
Pa = interp1(t_data, x.Pa, t, 'previous');
writematrix(x.Pa, 'signal.xlsx');
```

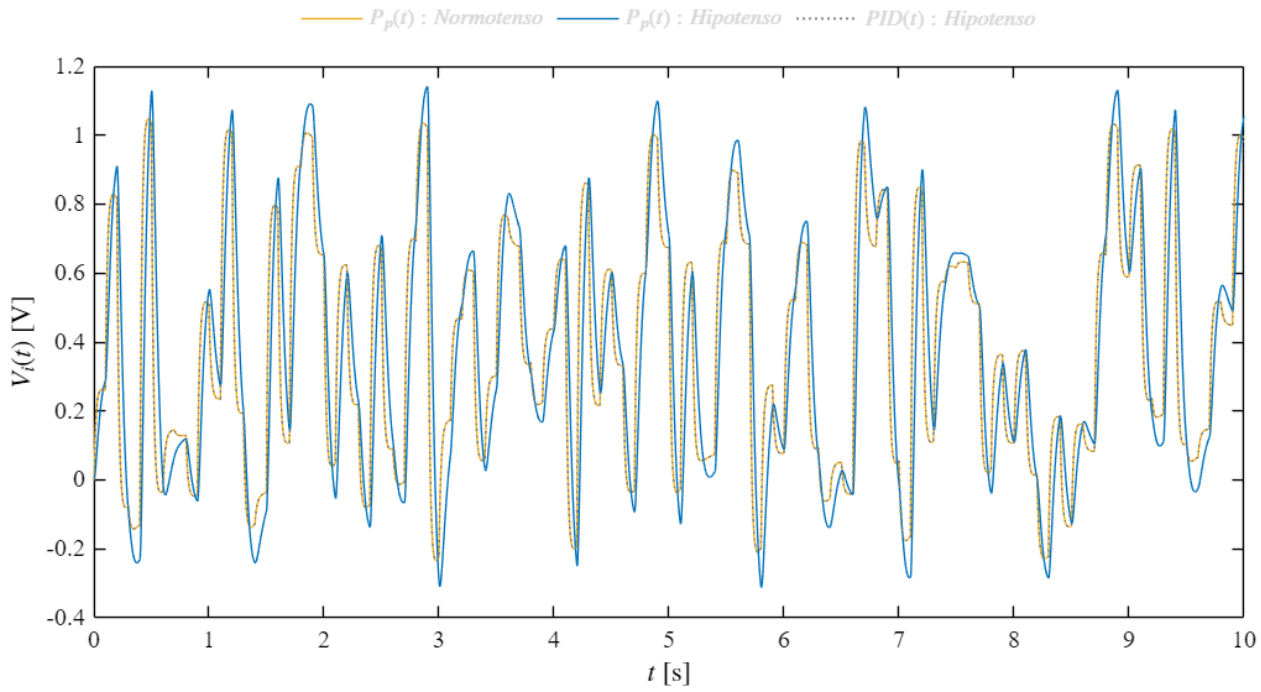
1. Respuestas: Hipotenso, Normotenso e Hipertenso

```
plotsignals(x.t,x.PpL,x.PpN,x.PpH,'individuos')
```



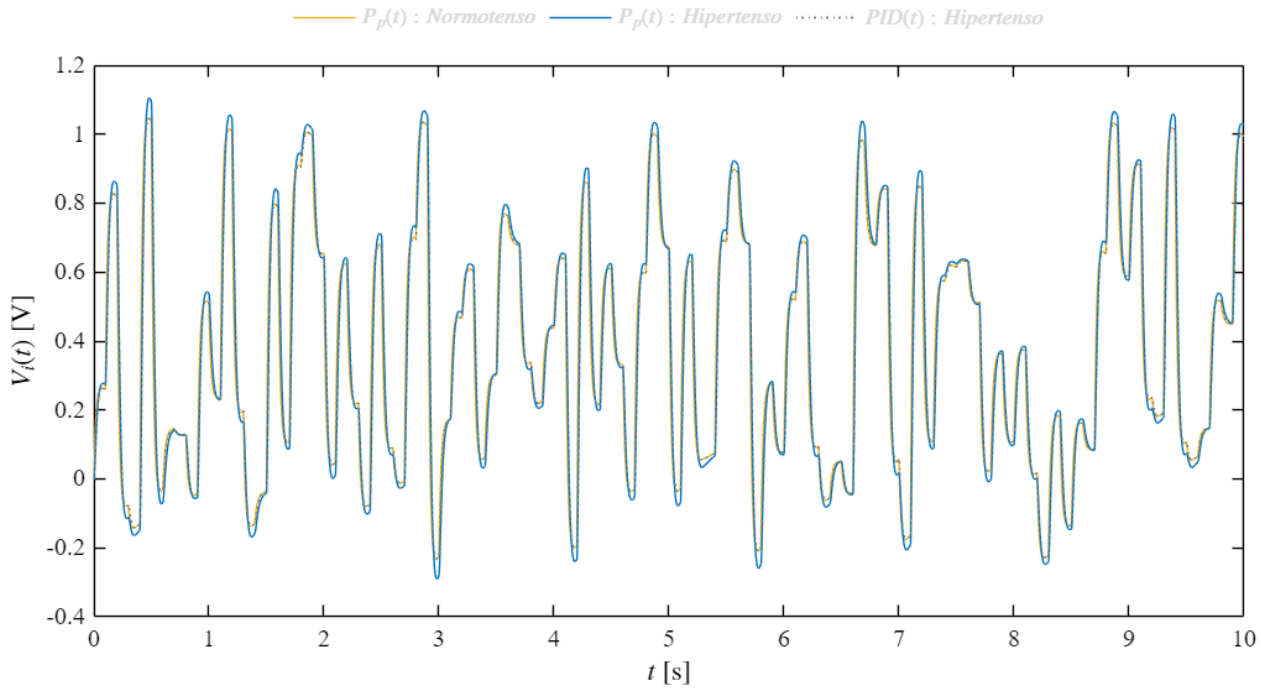
2. Respuestas: Normotenso, Hipotenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpL,x.PIDL,'Hipotenso')
```



3. Respuestas: Normotenso, Hipertenso y Controlador

```
plotsignals(x.t,x.PpN,x.PpH,x.PIDH,'Hipertenso')
```



Modelo de EDOs

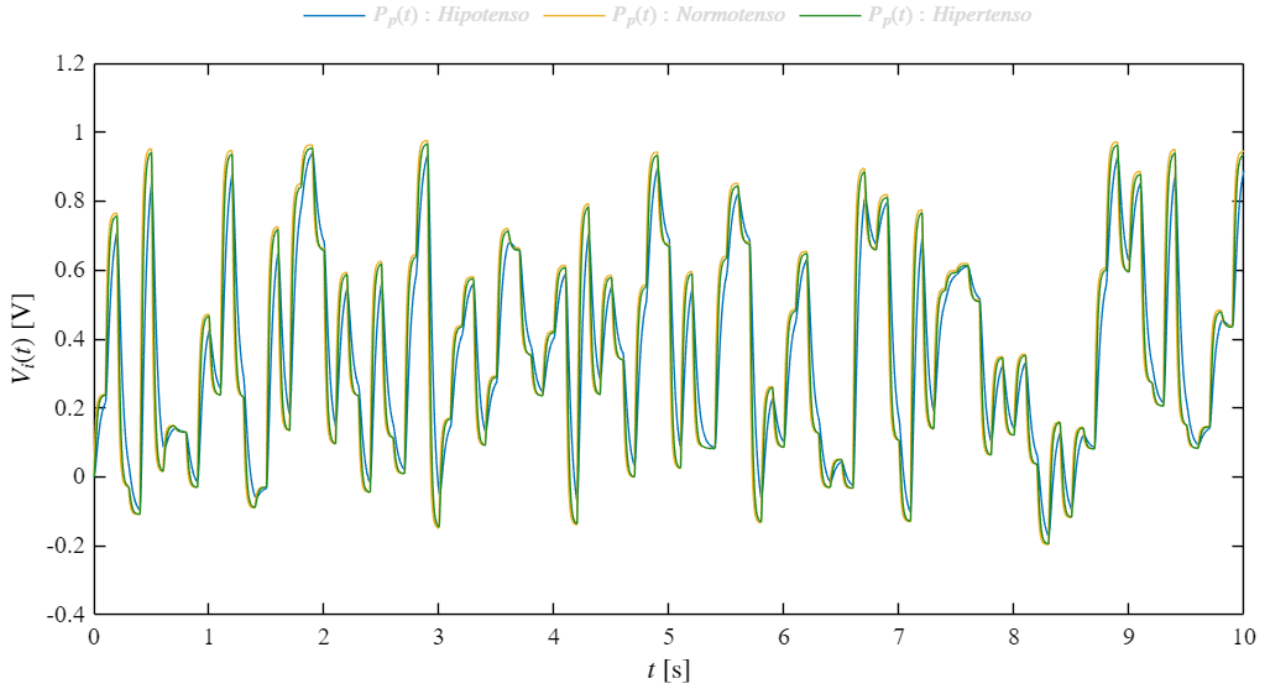
```
par_hipo = [0.01,0.001,1.00,3.500]; %[Z,L,R,C]
par_normo = [0.01,0.001,1.00,1.500];
```

```
par_normo = 1×4
    0.0100    0.0010    1.0000    1.5000
```

```
par_hiper = [0.01,0.001,1.00,1.000];
```

```
par_hiper = 1×4
    0.0100    0.0010    1.0000    1.0000
```

```
PpN = cardio(par_normo,Pa,t);
PpL = cardio(par_hipo,PpN,t);
PpH = cardio(par_hiper,PpN,t);
plotsignals(t,PpL,PpN,PpH,'odes')
```




```
function Pp = cardio(par,Pa,t)
    Z = par(1); L = par(2); R = par(3); C = par(4); dt = t(2);
    n = round(t(end)/dt);
    FL = zeros(length(t),1); FL(1) = 0;
    Pp = zeros(length(t),1); Pp(1) = 0;
    for i = 1 :n
        Fl(i+1) = FL(i) + (-Pp(i) + Pa(i)/L)*dt;
        Pp(i+1) = Pp(i) + (FL(i)/C - (1/(Z*C) + 1/(R*C))*Pp(i) + Pa(i)/(Z*C))*dt;
    end
end
```

Funcion: Respuestas de simulacion

```
function plotsignals(t,x1,x2,x3,signal)
set(figure(),'color','w')
set(gcf,'units','inches','position',[1,1,10,5])
set(gca,'fontname','times new
roman','fontsize',12,'color','w','xcolor','k','ycolor','k')
```

```
hold on; grid off; box on;
clr1 = [0,114,189]/255;
clr2 = [237,177,32]/255;
clr3 = [34,139,34]/255;
clr4 = [0.5, 0.5, 0.5];
order = [clr1;clr2;clr3;clr4]; colororder(order)

if signal == "individuos"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr1) %Hipotenso
    plot(t,x2,'-','LineWidth',0.25,'Color',clr2) %Normotenso
    plot(t,x3,'-','LineWidth',0.25,'Color',clr3) %Hipertenso
    L = legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');
elseif signal == "Hipotenso"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr2) %Normotenso
    plot(t,x2,'-','LineWidth',0.25,'Color',clr1) % Hipotenso
    plot(t,x3,':','LineWidth',1,'Color',clr4) %Controlador Hipotenso
    L = legend('$P_p(t):Normotenso$','$P_p(t):Hipotenso$','$PID(t):Hipotenso$');
elseif signal == "Hipertenso"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr2) %Normotenso
    plot(t,x2,'-','LineWidth',0.25,'Color',clr1) % Hipotenso
    plot(t,x3,':','LineWidth',1,'Color',clr4) %Controlador Hipertenso
    L = legend('$P_p(t):Normotenso$','$P_p(t):Hipertenso$','$PID(t):Hipertenso$');
elseif signal == "odes"
    plot(t,x1,'-','LineWidth',0.25,'Color',clr1) %Hipotenso
    plot(t,x2,'-','LineWidth',0.25,'Color',clr2) %Normotenso
    plot(t,x3,'-','LineWidth',0.25,'Color',clr3) %Hipertenso
    L = legend('$P_p(t):Hipotenso$','$P_p(t):Normotenso$','$P_p(t):Hipertenso$');
end

set(L,'interpreter','latex','location','northoutside','orientation','horizontal','bo
x','off')

xlabel('$t$ [s]','interpreter','latex')
ylabel('$V_i(t)$  [V]','interpreter','latex')
xlim([0 10]); xticks(0:1:10)
ylim([-0.4 1.2]); yticks(-0.4:0.2:1.2)

exportgraphics(gcf,[signal,'matlab.pdf'],'ContentType','vector')
end
```